# Sign Language Recognition

## Impact of Face Swapping & Data Augmentation

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## Vision Mamba for Pose-Based Static SLR

Shreyan Naskar

TCS Research - Intern

# Impact of Face Swapping & Data Augmentation on Sign Language Recognition

Bridging the communication gap between deaf and hearing communities through advanced Isolated Sign Language Recognition, introducing **CALSE-1000**, deepfake-based augmentation, and empirical validation via I3D neural networks.

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# The Global Imperative for Sign Language Recognition

## 430M

### With Hearing Loss

WHO-reported cases of moderate or higher severity hearing loss worldwide

## 1.5B

### Some Degree of Loss

Total individuals globally with any level of hearing impairment

## 300+

### Sign Languages

Distinct sign languages exist worldwide , yet only **1%** of the global population understands any of them

This communication barrier severely restricts deaf individuals' access to education, healthcare, and employment , making automated SLR a critical accessibility technology.

# Understanding Sign Language Recognition

### Visual Complexity

Sign languages are entirely visual, relying on **manual gestures** and **non-manual features** , facial expressions and body posture , to convey full grammatical meaning. They have independent grammar with no direct word-for-word correspondence to spoken languages.

## Two Recognition Paradigms

### CSLR

Continuous Sign Language Recognition , full sentence interpretation from video

### ISLR ✓

Isolated Sign Language Recognition , word-level gloss translation. **This study's focus.**

# The Data Scarcity & Privacy Problem

## Rigorous Recording Requirements

Controlled lighting, multiple viewpoints, and zero occlusions are mandatory , creating a high production barrier for every new dataset sample.

## Expert Validation Bottleneck

Because SL is opaque to most people, professional interpreters must supervise every recording session to certify accuracy , a costly and scarce resource.

## The Anonymization Dilemma

Pixelation, blackening, and greyscale , standard anonymization tools , **destroy facial features entirely**, rendering video useless for deep learning.

## Generalization Failures

Existing datasets lack signer variability. Models trained on homogeneous data fail to generalize to new users in real-world scenarios.

# Introducing CALSE-1000

**Conjunto Aislado de Lengua de Signos Española** , the largest publicly available dataset for Spanish Isolated Sign Language Recognition to date.

### 5,000 Videos

Total recordings across all glosses

### 1,000 Glosses

Distinct Spanish sign vocabulary entries

### ~15 Signers

Average signer diversity per gloss

## Data Aggregated From Three Sources

A professional LSE interpreter filled vocabulary gaps by recording missing glosses plus 3 additional samples per sign.

01

### DILSE

Dictionary of Spanish Sign Language

02

### STS

Spread the Sign multilingual dictionaries

03

### SACU

University of Seville community assistance dataset

# Ensuring Model Robustness: Multi-Scenario Recording

To prevent overfitting to a single signer's appearance, the collaborating interpreter was recorded under three deliberate stylistic variations:

1

### Scenario A

Face forward · hair tied · **high-intensity** signing emphasis

2

### Scenario B

Face forward · hair down · **natural everyday** signing emphasis

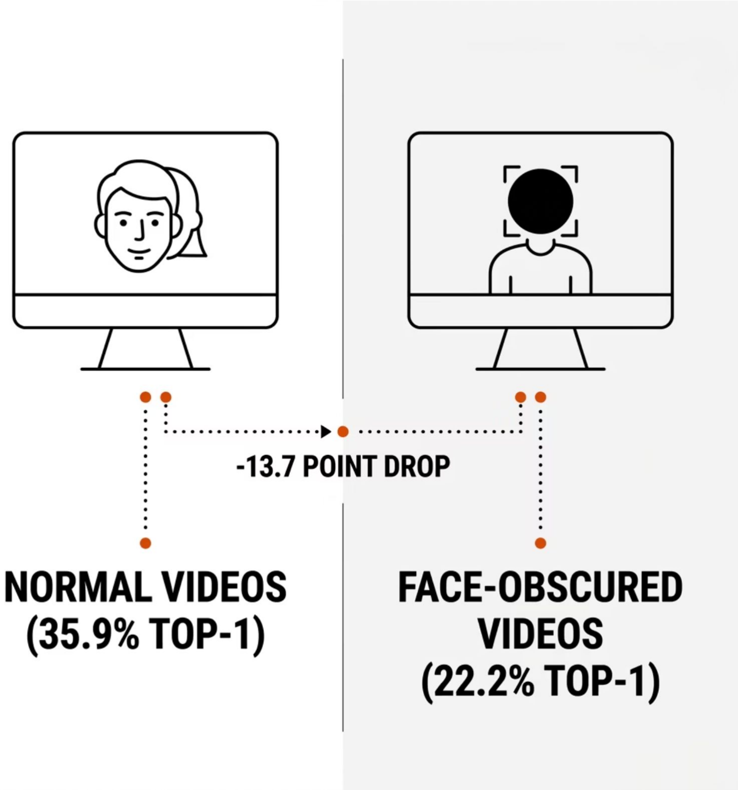
3

### Scenario C

**Profile perspective** · neutral emphasis · varied angle

This deliberate diversity in angle, style, and intensity forces the model to learn the sign itself , not the signer's background or appearance.

# Proving Facial Expressions Are Indispensable



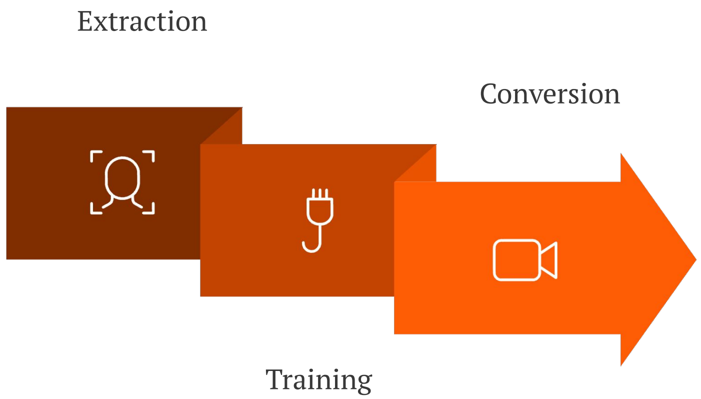

## The Facial Omission Experiment

Before applying face swapping, researchers needed empirical proof that facial data is genuinely indispensable , not merely assumed to be.

Using the *Face-parsing* PyTorch algorithm, signers' faces were segmented and completely blacked out across the DILSE test set.

**Result:** Top-1 accuracy collapsed from **35.9% → 22.2%** , a **13.7 point drop** , definitively proving simple pixelation is an unacceptable anonymization strategy for SLR.

# Face Swapping & Affine Transformations: The Augmentation Pipeline




Three synthetic identities (two male, one female) were generated using the open-source *FaceSwap* tool with the *Phaze-a* deep learning model. In parallel, affine transformations , scaling, translation, reflection, and rotation , were applied via the *RodGal-2020* library to teach the model spatial invariance.

# Experimental Design & Test Battery

## Cross-Dataset Evaluation Strategy

Given extreme data scarcity (~6 videos per sign), a traditional validation split would cripple training. The team used a **cross-dataset approach** with a 5:1 train-to-test ratio, rotating SACU, STS, and DILSE as the isolated test set across three experimental runs.

## I3D Network Configuration

- **Optimizer:** Adam · LR: $10^{-3}$ · Batch: 10
- **Epochs:** 60 (loss stabilization point)
- **Architecture:** Inflated 3D ConvNet (PyTorch)

## Augmentation Conditions

### Baseline

500 original training videos

### FS1

+200 face-swapped videos (700 total)

### AF1 / AF2

+200 affine-transformed videos (interpreter / public data)

### FS1-AF1 / FS1-AF2

Combined methods (900 total)

# Results, Key Findings & Future Directions

## 🏆 Best Result

DILSE test set: FS1 alone pushed Top-1 from **35.9% → 47.6%** , an **+11.7 point** gain (+32.59% relative improvement)

## ⚠️ Over-Augmentation Risk

Combining FS1+AF1 on the same signer's videos caused **overfitting** , accuracy dropped back to 37.8%. Isolated augmentation generalizes better.

## 🔒 Privacy Solved

Face swapping protects signer identity *while preserving* non-manual facial cues , overcoming the fundamental anonymization dilemma in SLR datasets.

## 🚀 Future Work

With CALSE-1000 established, the team will explore **advanced neural architectures** beyond I3D and aim to reduce the computational cost of scalable face swapping.

Top-5 accuracy improved by up to **+8.8 points** and Top-10 by **+9 points** , demonstrating consistent, broad-spectrum gains across all evaluation tiers.

# Vision Mamba for Pose-Based Static Sign Language Recognition

A Cross-Lingual Deep Learning Approach integrating MediaPipe landmark extraction with state-of-the-art vision architectures , benchmarking ConvNeXt, Swin Transformer, and Vision Mamba across Turkish, Arabic, and American Sign Languages.

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# Research Scope and Novelty

This study deliberately focuses on **isolated static sign and fingerspelling recognition** , establishing a lightweight, low-latency baseline essential for embedded devices, edge computing, and real-time educational tools.

### 🌐 Cross-Lingual Coverage

Turkish (TSL), Arabic (ArSL), and American (ASL) Sign Languages , three structurally diverse linguistic families unified under one framework.

### 🏗️ Architectural Benchmarking

First rigorous comparison of ConvNeXt, Swin Transformer, and Vision Mamba on identical 2D pose features , isolating architecture from dataset bias.

### ⚡ Real-Time Viability

End-to-end pipeline optimized for high FPS and low latency , validated against real-world spontaneous signing conditions.

# Research Gaps and Prior Work

The existing literature, while extensive, leaves three critical gaps that this work directly targets.

### Gap 1: Sensor Dependency & Hardware Barriers

Foundational SLR datasets (e.g., AUTSL) required Kinect v2 sensors, Leap Motion, or data gloves, inherently limiting scalability. This framework uses only standard RGB camera input for mass accessibility.

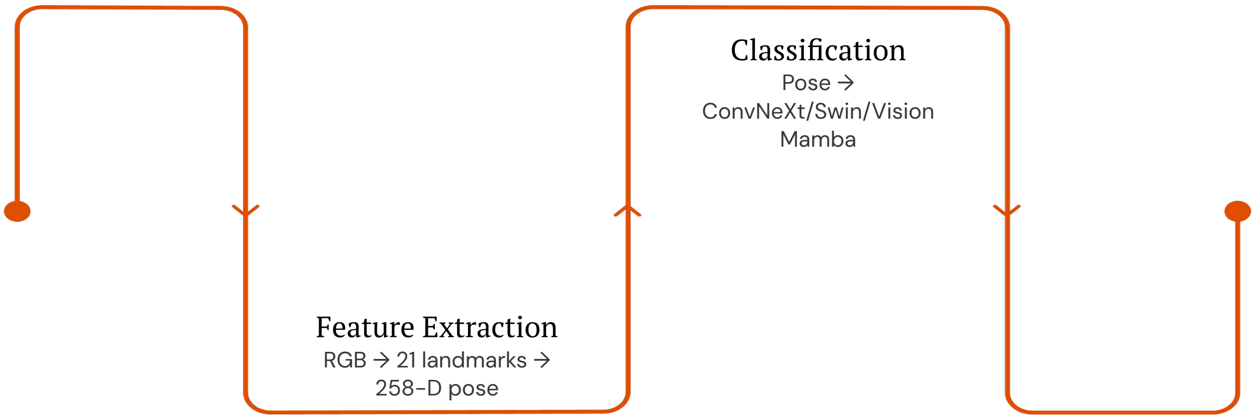
### Gap 2: Single-Language Silos

Most published systems are rigidly single-source. No prior work rigorously benchmarks the latest architectures, SSMs, modern CNNs, and ViTs, against each other across multiple sign languages with an identical feature set.

### Gap 3: Raw Pixel Dependency

Existing pipelines (e.g., YOLO + LSTM) incur two-stage computational overhead and process raw image pixels rather than compact, robust 2D pose representations, reducing real-time efficiency and generalization.

# System Pipeline Architecture




The pipeline intentionally decouples feature extraction from classification. MediaPipe distills each raw RGB frame into a 258-dimensional, scale-invariant pose vector , purifying downstream architectural comparisons by eliminating background clutter and image noise as confounding variables.

## Stage 1: MediaPipe Hands

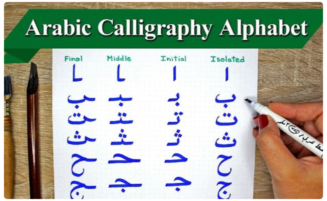
- 21 anatomical landmarks per hand (wrist, CMC, MCP, PIP, DIP, TIP joints)
- Inherently normalized to bounding box , scale and translation invariant
- Outputs a compact 258-feature coordinate vector

## Stage 2: Architecture Trio

- Three independent models trained per language (9 total models)
- Identical input vectors ensure a controlled, fair comparison
- AdamW / Adam optimizers, cosine annealing, 50 epochs, 5-fold CV

# Dataset Construction: 130,000+ Curated Images

Data was aggregated from **9 distinct public-domain repositories** , ensuring exposure to diverse hand sizes, genders, lighting conditions, and camera distances. Exactly **1,500 samples per class** were retained after curation and augmentation (flipping, minor pixel replacement). Major rotations were strictly excluded to preserve pose-label validity.

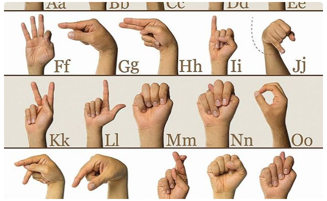



## Arabic SLR (ArSL)

**32 classes** , 28 Arabic letters + 4 special combined signs. Sources: KArSL, ASLAD-190K, ArSL21L, RGB ArSL.


## Turkish SLR (TSL)

**29 classes** , full Turkish alphabet. Sources: Tr SL Dataset + Spreadthesign TSL (standardized, officially recognized signs).




## American SLR (ASL)

**26 classes** , English alphabet. Sources: ASL Mendeley Data, ASLYset, SignAlphaSet , covering diverse signers, resolutions, and perspectives.

# Architectural Candidates: Three Paradigms

### ConvNeXt-Tiny

Modernized CNN with Transformer-inspired design: 7×7 kernels, inverted bottlenecks, layer normalization. Exceptional local feature extraction with high throughput. Retains CNN inductive bias.

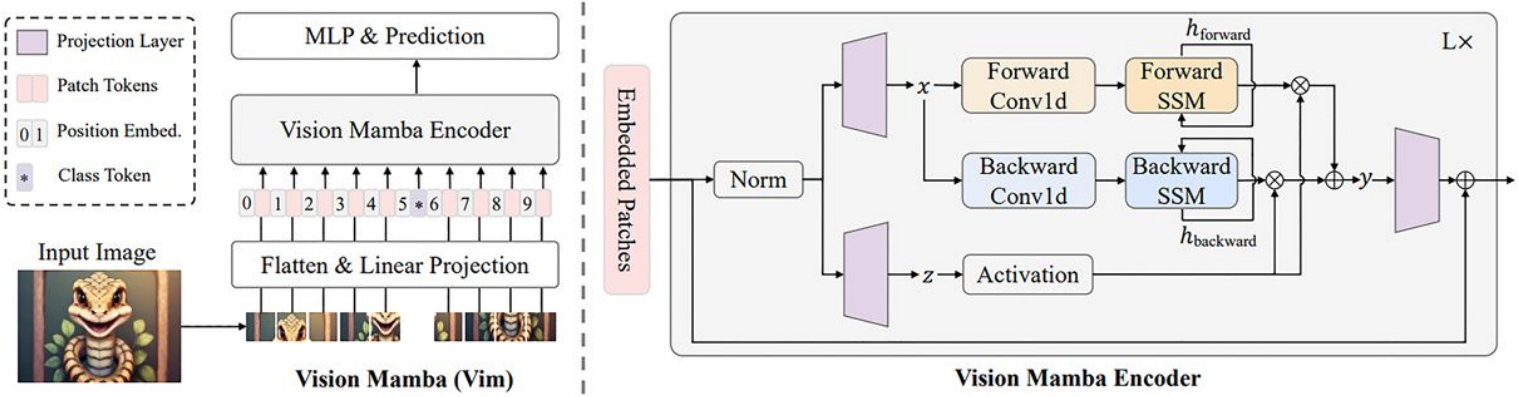
### Swin Transformer-Tiny

Hierarchical ViT with shifted-window attention reducing quadratic complexity to linear. Captures non-local relationships while confining attention to local windows for efficiency.

### Vision Mamba-Tiny

State Space Model replacing self-attention with Visual State Space (VSS) blocks. Selective Scan Module dynamically retains or forgets information , achieving true linear complexity across sequence length.

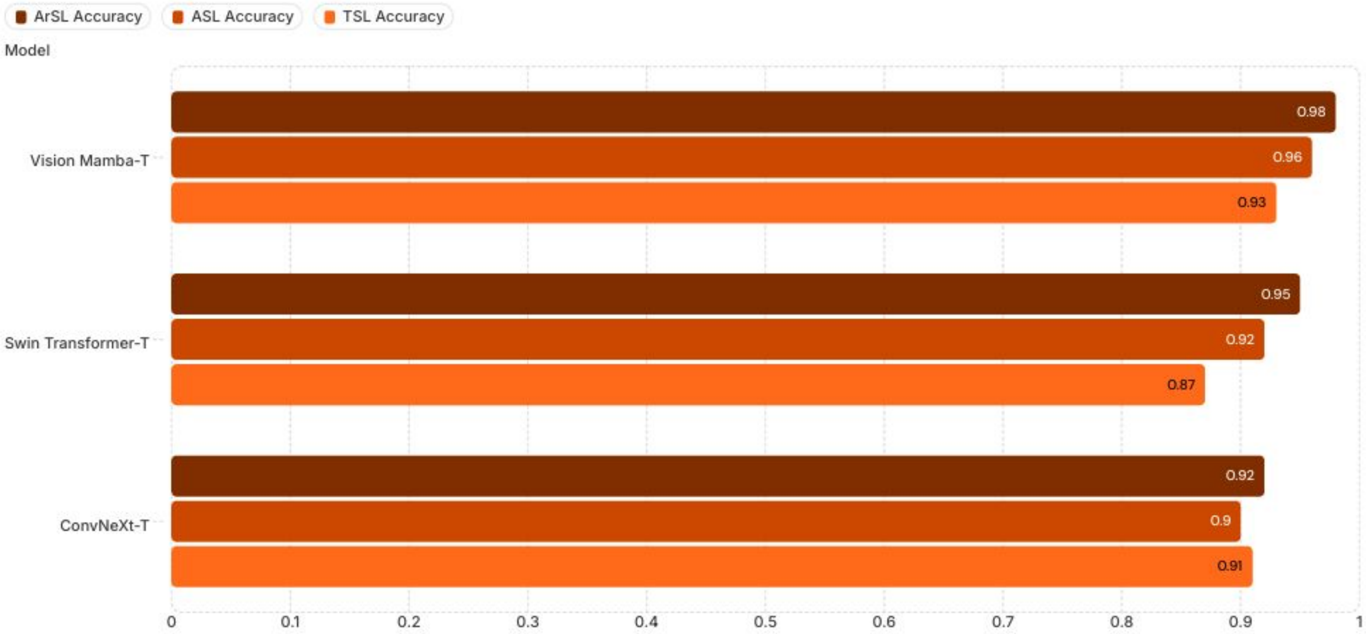
# Vision MAMBA




- New vision architecture designed for efficient and scalable visual representation learning.
- Inspired by State Space Models (SSMs) replacing self-attention with the Visual State Space (VSS) block, enabling efficient modeling of long-range dependencies in images.
- Selective state space mechanism dynamically focuses on important visual information while ignoring irrelevant details.
- Achieves **linear** computational complexity with respect to sequence length.
- Highly memory-efficient and scalable, especially for high-resolution images.

# Experimental Results: Accuracy Across All Languages

All models trained with pre-trained ImageNet weights, AdamW/Adam optimizers, learning rate $1\times10^{-4}$ with cosine annealing, 50 epochs, batch size 32, and 5-fold cross-validation. Vision Mamba achieved top-tier performance universally — its Selective Scan Module efficiently distinguishing highly similar signs across all three linguistic families.




Notable exception: ConvNeXt outperformed Swin on TSL (0.91 vs. 0.87) , its spatially localized inductive bias proved uniquely effective for TSL's complex two-handed configurations.

# Real-World Validation and Throughput

## GUI User Study

Ten diverse participants tested **300 real-time spontaneous sign instances** through a custom interface featuring live video feeds, bounding boxes, text output management, and rapid language-toggling. Result: **94.2% macro accuracy** , closely matching theoretical test-set metrics.

## Cross-Dataset Generalization

Tested on entirely unseen **LSA16 (Argentinian SL)** and **RWTH (German SL)** datasets without retraining. Vmamba maintained **70–98% accuracy** on equivalent character signs, definitively proving it learns transferable semantic pose features , not signer-specific memorization.

## Inference Performance (NVIDIA RTX 3080)

| Model | FPS | Latency | Size |
|---|---|---|---|
| Vision Mamba-T | **67.5** | **14.8 ms** | 334.5 MB |
| ConvNeXt-T | 64.5 | 15.5 ms | 106 MB |
| Swin-T | 39.8 | 25.1 ms | , |

Vmamba achieved the highest FPS and lowest latency despite its larger footprint , its linear-scaling SSM is optimal for real-time edge inference.

# Conclusions and Forward Trajectory

**Vision Mamba is unequivocally established as the superior architecture** for pose-based static SLR , delivering the best combined profile of F1-score, throughput (67.5 FPS), and latency (14.8 ms) across all three sign languages.

## Architectural Supremacy Confirmed

Vmamba's Selective Scan Module captures long-range dependencies across sparse 2D keypoint sequences with linear complexity , outperforming both attention-based and convolutional paradigms on abstract geometric pose data.

## Methodological Rigor Validated

Independent per-language training successfully isolated intrinsic architectural bias from linguistic data complexity, enabling precise attribution of performance differences to model design vs. sign complexity (e.g., TSL two-handed signs).

## Future Work: Unified Multilingual Vmamba

The foundation is laid to pursue a single multilingual model with shared encoder weights and joint training across all three languages , alongside further optimization for low-power edge-computing and mobile deployment platforms.